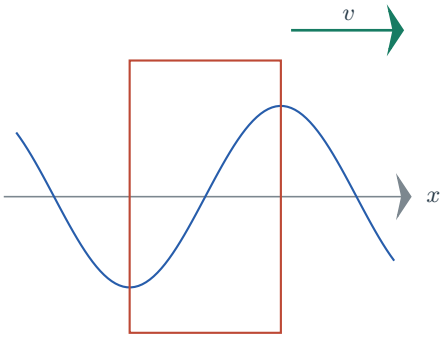
M5-154 矩形回路在行波磁场中的平均推力



红框整体平移，蓝线表示固定时刻的 $B_z(x)$

$$\Omega = \omega - kv$$

$$\mathcal{E}_0 = 2B_0 h(\Omega/k)\sin(k\ell/2)$$

$$\langle F_x \rangle = \frac{2B_0^2 h^2 \Omega}{kR}\sin^2(k\ell/2)$$